

Time synchronisation over 10BASE-T1S: software stamps, LAN8651 hardware stamps, a PI clock servo

Generated by `make_ptp_report.py` from the console logs in `docs/t1s_results/sync`. Bench: ESP32-S3 + LAN8651 HAT (node, PLCA coordinator, SPI 20 MHz) =T1S= media converter –100BASE-TX– ESP32-S3 + W5500 (ESP-B). 2026-10-06.

1. Measuring: two-way time transfer

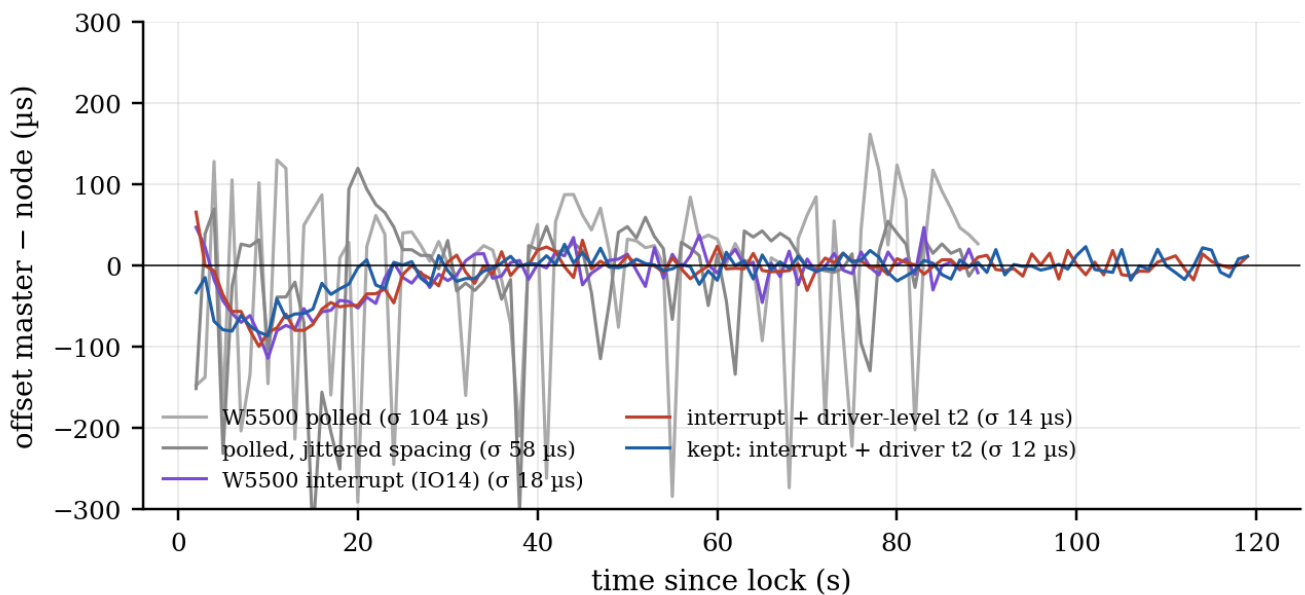
ESP-B asks, the node answers (UDP 5007): $t1..t4$, $\text{offset} = ((t2 - t1) + (t3 - t4)) / 2$. Precision is the scatter of the offset around its drift line; with no reference clock on the bench, accuracy is not measured.

stamps	exchanges	σ , all	σ , min-delay per 2 s	drift
software, both boards	998	147 μs	53 μs	-1.7 ppm
node hardware, ESP-B software	876	169 μs	15 μs	+19.8 ppm

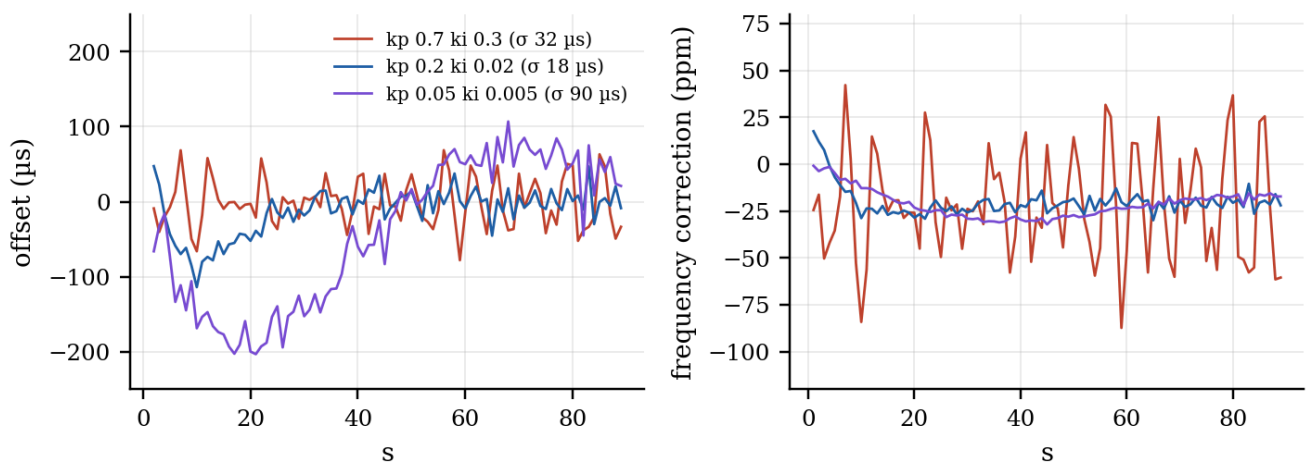
The node's hardware stamps come from the LAN8651's 1588 wall clock (TSU, 40 ns per 25 MHz tick). Two settings had to be found: `OA_CONFIG0.FTSE/FTSS` take effect only when written with `SYNC` at init, and the PHY's packet matcher, which triggers the stamp at the SFD on the wire, is off out of reset (`STATUS1.TTSCMA`: capture requested, not triggered). The drift differs between the rows because the hardware clock runs off the HAT's crystal, the software one off the ESP32's.

2. Steering: a PI servo on the LAN8651 clock (`ptp lock`)

The node asks ESP-B 16 times a second (its request and the reply stamped in hardware), keeps the min-delay exchange per second, steps once (TA register below 1 s, else TSL/TN) and then steers the TSU frequency ($TI + TISUBN$, 2^{-24} ns steps). Offset after the first 20 s:



run	master side	kp / ki	σ	max	mean	one-way delay p50	frequency held
lock_0.7_0.3	W5500 polled	0.7 / 0.3	233.7 μs	595 μ s	-0.2 μ s	802 μ s	-15.7 $\pm$ 157.7 ppm
lock_0.2_0.02	W5500 polled	0.2 / 0.02	104.3 μs	291 μ s	-4.5 μ s	1019 μ s	-21.7 $\pm$ 21.9 ppm
lock_0.05_0.005	W5500 polled	0.05 / 0.005	116.8 μs	282 μ s	-49.4 μ s	970 μ s	-24.8 $\pm$ 5.9 ppm
lock1	W5500 polled (first run)	0.7 / 0.3	38.1 μs	90 μ s	+0.1 μ s	750 μ s	-20.3 $\pm$ 33.7 ppm
lock2s_0.4_0.08	+ post-send t3 (reverted)	0.4 / 0.08	38.3 μs	240 μ s	+0.3 μ s	197 μ s	-20.9 $\pm$ 17.4 ppm
lock2s_0.2_0.02	+ post-send t3 (reverted)	0.2 / 0.02	23.1 μs	78 μ s	-2.0 μ s	193 μ s	-21.3 $\pm$ 4.8 ppm
lockdrv_0.4_0.08	interrupt + driver-level t2	0.4 / 0.08	14.6 μs	37 μ s	-0.2 μ s	533 μ s	-21.0 $\pm$ 6.6 ppm
lockdrv_0.2_0.02	interrupt + driver-level t2	0.2 / 0.02	13.9 μs	49 μ s	-1.8 μ s	532 μ s	-21.4 $\pm$ 2.7 ppm
lockfinal_0	kept: interrupt + driver t2	0.2 / 0.02	14.9 μs	48 μ s	-1.0 μ s	520 μ s	-21.0 $\pm$ 3.1 ppm
lockfinal_1	kept: interrupt + driver t2	0.2 / 0.02	11.6 μs	28 μ s	-0.3 μ s	519 μ s	-20.8 $\pm$ 2.5 ppm
lockint_0.7_0.3	W5500 interrupt (IO14)	0.7 / 0.3	32.2 μs	78 μ s	-0.7 μ s	740 μ s	-21.3 $\pm$ 28.5 ppm
lockint_0.2_0.02	W5500 interrupt (IO14)	0.2 / 0.02	18.5 μs	52 μ s	-2.3 μ s	736 μ s	-21.4 $\pm$ 3.7 ppm
lockint_0.05_0.005	W5500 interrupt (IO14)	0.05 / 0.005	89.8 μs	203 μ s	-17.4 μ s	741 μ s	-24.0 $\pm$ 4.9 ppm
lockj_0.7_0.3	polled, jittered spacing	0.7 / 0.3	200.7 μs	436 μ s	+2.4 μ s	851 μ s	-15.1 $\pm$ 134.8 ppm
lockj_0.2_0.02	polled, jittered spacing	0.2 / 0.02	58.1 μs	301 μ s	+3.7 μ s	1005 μ s	-19.9 $\pm$ 12.3 ppm
lockj_0.05_0.005	polled, jittered spacing	0.05 / 0.005	94.4 μs	190 μ s	-19.7 μ s	951 μ s	-25.2 $\pm$ 4.9 ppm



Reading it. The servo locks within 2 s and holds about -21 ppm, the HAT crystal against the ESP32's. What limits the result is the master's side, step by step: (1) ESP-B's W5500 was polled every 1 ms, so its receive time wandered by up to 1 ms -- the Elite wires the W5500's INTn to IO14, and interrupt-driven receive halves the scatter; (2) ESP-B stamped t2 in the UDP socket task -- a tap in the Ethernet driver's receive path, before lwIP, takes another quarter off; (3) a t3 taken after the send returned (a software two-step) scattered more than one taken just before it and was reverted. A request spacing of whole milliseconds kept every request at the same poll phase, so the spacing is jittered by 0–1 ms. ptp4l's gains (0.7 / 0.3) turn this bench's measurement noise into ± 30 –40 ppm frequency swings; 0.2 / 0.02 is the default.

3. Limits

- One side only has hardware stamps; the master is software (ESP-B) and the path crosses a store-and-forward converter whose queueing is not symmetric. Two LAN8651 nodes (hardware at both ends, no converter) are the next measurement; the servo already uses a master's hardware follow-up when it gets one.
- No reference: offsets are precision, not accuracy. A PPS on DIOA0 from each node and an oscilloscope would give accuracy directly (the firmware has ptp_pps; Rev C ties DIOA to ground).
- Timestamps on every received frame cost 6 % of receive throughput ($8.75 \rightarrow 8.20$ Mbit/s at 1472 B), so they are a build option (`-DLAN865X_FRAME_TIMESTAMPS`).